

MAYUR DOKRAS

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Professional Summary

Software Engineer with over 4 years of experience and an MSc in Software Engineering from the University of Southampton, specializing in secure systems and binary level reverse engineering. I have a strong background in analyzing MCUs and ECUs, specifically using Ghidra and Binary Ninja to decompile binaries, extract keys, and mitigate vulnerabilities like Buffer Overflow and software hijacking. During my time at Capgemini on a CISCO project, I focused on Programmable I/O Peripherals, where I reverse engineered embedded systems and developed Python and C++ integrations to enhance performance. I am highly proficient in C/C++ and SQL/PL-SQL, with a deep understanding of the full SDLC and a commitment to maintaining ISO 26262 safety standards. Having worked with globally distributed teams, I am fully equipped to solve intricate technical challenges in a high-performance remote environment.

Relevant Work Experience

Backend Developer – Malvox

University of Southampton

February 2025 - July 2025

- Designed relational data models and built SQL driven backend workflows for a cybersecurity platform integrating UK Government APIs.
- Developed backend logic for authentication and access control, with secure handling of sensitive data using AES-256 encryption and key level protection for critical records.
- Strengthened API to database security by adding input validation, parameterized queries and defensive checks to prevent SQL injection attacks.
- Improved reliability of backend operations through transactional updates, structured error handling and close collaboration with cross-functional teams to deliver secure features across Agile sprints.

Associate II - Software Engineer

Capgemini Engineering

November 2021 - November 2024

- Worked hands on in C/C++ and Linux on low-level software used in TOR setups within GPON environments, including CCM/CFM protocol development, debugging and performance focused improvements across major deployments.
- Built Python based backend tooling around these workflows to detect client side faults faster, with structured logs, clear error codes and severity levels integrated into existing C++ call flows.
- Delivered 7+ features and resolved 40+ production defects using TDD, pair programming and Agile/Scrum practices in day to day engineering.
- Used CI/CD (Jenkins), Git, GDB, Wireshark and monitoring to maintain release stability, reduce MTTR and improve production reliability of backend connected systems.

Freelance Mobile Application Developer

Self Employed

February 2021 - October 2021

- Built Flutter based mobile apps including e-commerce with admin, customer, delivery boy apps, Grocery Apps (admin, customer) and Gym Workout Assistant app.
- Integrated payment gateways (Razorpay, PayPal) for e-commerce applications and Firebase as backend, authentication and offline data storage delivered end-to-end solutions. Evaluated PostgreSQL for e-commerce backend but chose NoSQL (Firebase) for faster delivery.

Education

MSc in Software Engineering

University of Southampton, UK – 2024/25

- Dissertation: Link2Feed full-stack Android app using Kotlin, Jetpack Compose, Firebase and UK Gov APIs.
- During Second Semester for Software Security, I did binary level reverse engineering and vulnerability research as part of software security coursework. I used Ghidra and Binary Ninja to decompile protected applications, successfully bypassing license check logic and identifying critical flaws. This involved deep-diving into assembly and binary code to understand and mitigate low level system exploits.
- Used MongoDB and Oracle (PL-SQL) in Semester 2 project for Backend application development assignments.
- Developing an iOS app using SwiftUI with shared Firebase backend and LLM integration.

Bachelor's in Information Technology (CGPA: 6.99)

AISSMS IOIT, India – 2018/20

- Developed Image Re-Ranking web application using algorithms and data structures (HashMap-based ranking), JavaScript, PHP, SQL resulted in a published research paper on topic-based image categorization.

Diploma in Information Technology (Percentage: 73.18%)

RIT Polytechnic, India – 2015/17

- Developed a Java-based steganography system using the LSB algorithm.

Skills

- **Core Languages:** C/C++, Python, Java, JavaScript, Kotlin, Swift, Dart
- **Backend & API:** REST API Development, Java (Spring Boot), Python, API-based frameworks, OpenAPI, API Design
- **Databases:** MySQL, SQL, MongoDB, Oracle (PL-SQL) (used in basic practice applications, willing to deepen knowledge), PostgreSQL (explored for e-commerce while freelancing, now eager to learn more), Firebase
- **Frontend:** React.js, TypeScript, HTML5, CSS3, Responsive UI, React Testing Library, Angular (willing to expand knowledge)
- **DevOps & Cloud:** CI/CD, Git, AWS, Agile Methodology (Scrum), Linux
- **Concepts:** Full SDLC, secure coding, unit testing, code reviews, application resiliency, data structures, distributed systems

Certifications

- Java Object-Oriented Programming (LinkedIn 2021)
- Amazon Web Service S3 Bucket (Coursera 2021)
- Agile Software Development (Coursera 2022)
- Dart Functions Framework (Google Cloud 2023)
- Cloud Computing (IBM 2023)
- Hands-On Essentials: Data Warehousing Workshop (Snowflake 2026)